A Systematic Comparison of Large and Small Language Models in Standalone, Hybrid & Multi-Agent Architectures

Doğukan Topçu · Abdulbaki Çeltik · Berkhan Kargılı · Meriç Kelemeden
Advisors: Dr. Hüseyin Ünlü · Prof. Dr. Onur Demirörs
İzmir Institute of Technology · İzmir, Türkiye · June 2026

1 Abstract / Summary

Measurement-first platform benchmarking **ten architectures in three families** across **five public benchmarks** under a unified observability contract (accuracy · latency · cost · energy · carbon). We introduce **EATS** (Efficiency-Accuracy Trade-off Score) as an architecture-agnostic metric for comparing quality and efficiency together. Routing reaches **EATS 0.893** on HellaSwag and **0.854** on TruthfulQA; Pure Swarm leads GSM8K at **EATS 0.841**; Speculative achieves **EATS 0.908** on ARC. Novel **Blackboard-style** and **swarm** designs show task-specific strengths on reasoning benchmarks.

2 Background / Literature Review

LLMs deliver state-of-the-art reasoning but impose high latency, cost, and energy. **SLMs** are cheaper and data-sovereign, yet trail on broad reasoning tasks.

A systematic literature review of **54 collaborative SLM–LLM studies** indicates that, while hybrid and multi-agent designs are becoming more common, **cost and energy are still rarely treated as primary evaluation outcomes** alongside accuracy.

RQ1	**Performance.** How do collaborative SLM/LLM systems compare with standalone LLMs on reasoning & classification?
RQ2	**Efficiency.** How do hybrid & multi-agent designs affect cost, latency, and energy?
RQ3	**Unified metric.** Does EATS give a stable, architecture-agnostic ranking of the quality–efficiency trade-off?
RQ4	**Empirical grounding.** Do measured comparisons corroborate the trends from our 54-study SLR?

3 Measurement-First Platform

Every architecture consumes the same `Query` and returns the same `Response` carrying **accuracy, latency, tokens, cost, energy & CO_2** per query. The architecture layer is the *only* interchangeable component.

Experiment Config
architecture · benchmark · seed · models

↓

Control Plane
FastAPI · Next.js UI · runner · host-lock

↓

Architecture Layer
routing · speculative · oracle · ensemble · swarm

↓

Metrics Proxy + vLLM
latency · GPU power (NVML) · energy · CO_2 · cost

↓

Scorer → MLflow + Reports
per-item & aggregate metrics

TIER	HOST	VRAM	MODELS
SLM	NVIDIA L4	24 GB	gemma4, qwen3.5, llama3.2 (3–4B)
Mid LLM	NVIDIA RTX PRO6000	96 GB	gpt-oss, gemma4, qwen3.5 (20–35B dense + MoE)
Heavy LLM	NVIDIA H200	141 GB	gpt-oss, llama3.3 (70–120B)

4 EATS · Efficiency–Accuracy Trade-off Score

A single score comparing any architecture against the standalone-LLM baseline.

$$\mathrm{EATS} = \frac{Accuracy}{Accuracy + \lambda \cdot P_{\mathrm{efficiency}} + \beta \cdot P_{\mathrm{accuracy}}}$$

$$P_{\mathrm{efficiency}} = 0.65 \cdot Cost + 0.20 \cdot Latency + 0.15 \cdot Energy$$

$$P_{\mathrm{accuracy}} = 1 - Accuracy$$

$\beta = 0.60$ $\lambda = 0.40$

all terms normalised per standalone-LLM baseline · $\beta = 0.60$, $\lambda = 0.40$ calibrated via grid search on MMLU validation split

5 Energy, Carbon & Cost

Per-request measurements via GPU power sampling (NVIDIA NVML) at the inference host.

$Cost = \text{GPU rate} \cdot \text{time}$ $E = P_{\mathrm{GPU}} \cdot \Delta t$

$Latency = \Delta t$ per request $CO_2 = k \cdot E$

$k = 380\ \mathrm{gCO_2/kWh}$ (EU avg) · E in J ÷ 3.6×10^6

6 Benchmarks & Protocol

Five benchmarks span the **capability dimensions most cited in our companion SLR**, letting us test whether efficiency gains persist as tasks shift from knowledge retrieval to chain-of-thought math and adversarial fact-checking.

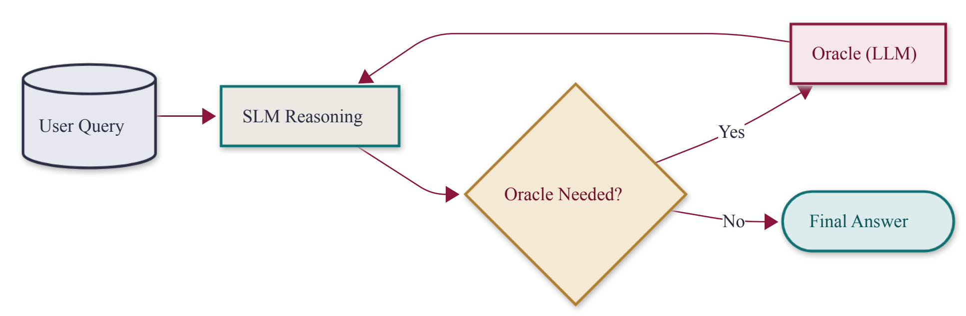
BENCHMARK	WHAT IT MEASURES
MMLU	**Broad knowledge** — 57 academic disciplines
ARC-C	**Scientific reasoning** — multi-step physical-world science
GSM8K	**Mathematical reasoning** — grade-school math, free-form
HellaSwag	**Commonsense inference** — sentence completion
TruthfulQA	**Adversarial truthfulness** — calibration vs. persuasive falsehoods

7 Architecture Taxonomy

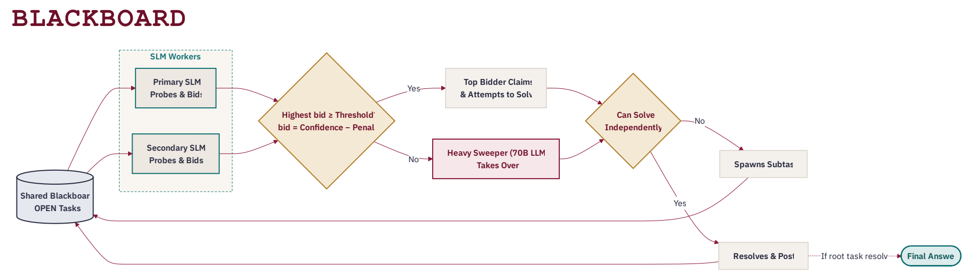
Ten architectures in three families. † marks novel project contributions introduced in the current study.

FAMILY	ARCHITECTURE	KEY MECHANISM
STAND	Single-Model	direct inference
STAND	MoE	internal sparse expert activation
HYBRID	Routing	confidence SLM→LLM escalation
HYBRID	Speculative	SLM drafts; LLM rewrites suffix
HYBRID	Debate (POA)	proponent–opponent–arbitrator
HYBRID	Active Oracle †	targeted LLM sub-queries
HYBRID	Blackboard †	bid-based task board; LLM sweeper fallback
HYBRID	Entropy Blackboard †	uncertainty-penalised bidding
MULTI	Ensemble	majority or confidence-weighted voting
MULTI	Pure Swarm †	SLM-only peer coordination; no LLM online

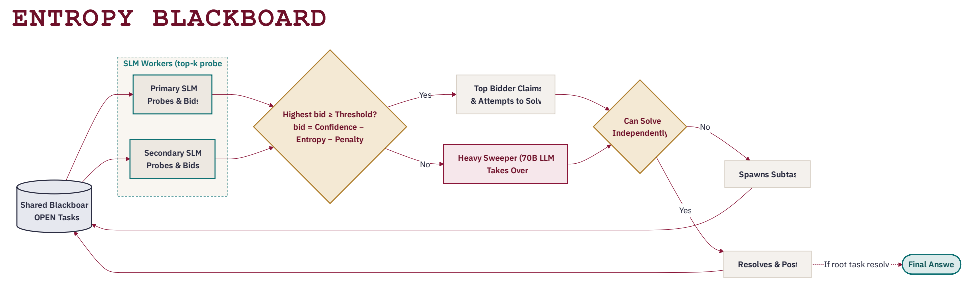
ACTIVE ORACLE



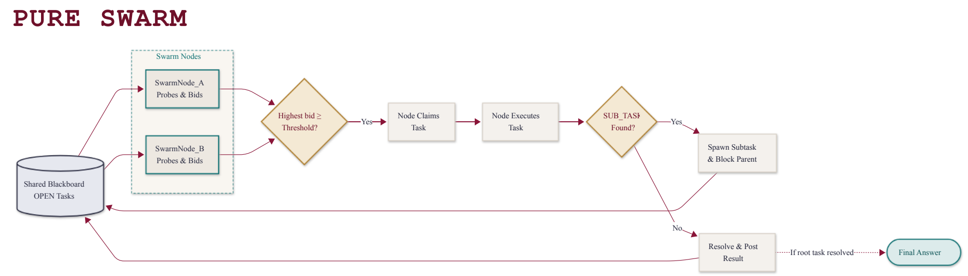
BLACKBOARD



ENTROPY BLACKBOARD



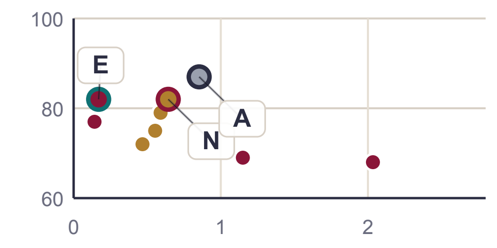
PURE SWARM



8 Benchmark Readouts

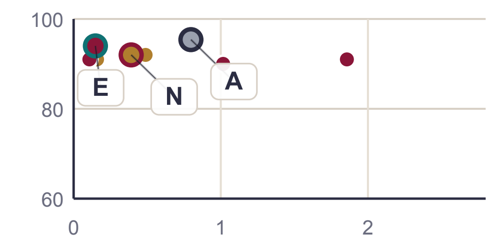
Each mini-plot shows accuracy against efficiency penalty, so the strongest trade-offs appear toward the upper-left. **E** = EATS winner, **A** = accuracy leader, **N** = best novel result.

MMLU



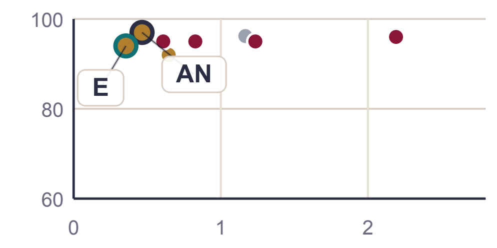
EATS winner Speculative
ACC leader gpt-oss-120b · 87.0%
NOVEL best Entropy BB · 82.0%

ARC



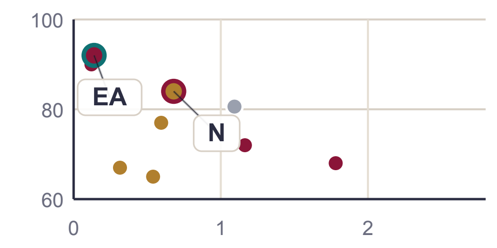
EATS winner Speculative
ACC leader gpt-oss-120b · 95.4%
NOVEL best Pure Swarm · 92.0%

GSM8K



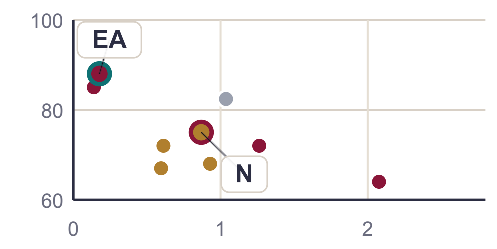
EATS winner Pure Swarm
ACC leader Entropy BB · 97.0%
NOVEL best Entropy BB · 97.0%

HellaSwag



EATS winner Speculative
ACC leader Speculative · 92.0%
NOVEL best Entropy BB · 84.0%

TruthfulQA



EATS winner Speculative
ACC leader Speculative · 88.0%
NOVEL best Entropy BB · 75.0%

9 EATS Landscape

Canonical EATS for the latest qualified run of each architecture; the standalone row shows the best MoE baseline on each benchmark.

ARCHITECTURE	MMLU	ARC	GSM8K	HSWAG	TQA
Routing	0.80	0.90	0.78	0.89	0.85
Speculative	**0.82**	**0.91**	0.72	**0.90**	**0.86**
Standalone (MoE)	0.68	0.73	0.66	0.59	0.61
Blackboard †	0.69	0.80	0.82	0.67	0.55
Entropy BB †	0.69	0.79	0.83	0.69	0.60
Active O. †	0.67	0.89	0.75	0.67	0.64
Debate POA	0.52	0.66	0.64	0.53	0.52
Pure Swarm †	0.67	0.82	**0.84**	0.60	0.61
Ensemble	0.40	0.53	0.52	0.43	0.38

Darker = higher EATS · bold = benchmark winner · tinted rows / † = novel contribution.

10 Conclusions

RQ1	Collaborative systems are competitive with standalone LLMs; the strongest results come from routing, speculative decoding, and novel coordination on reasoning-heavy tasks.
RQ2	Hybrid escalation gives the clearest cost and energy gains, while multi-agent coordination usually pays a latency penalty.
RQ3	EATS gives a consistent cross-architecture ranking: speculative leads four benchmarks, and Pure Swarm leads GSM8K.
RQ4	The measurements corroborate 8 of 10 SLR trends: SLM-first escalation is the strongest general pattern, while coordination and entropy-aware bidding help on task-specific cases.

⚠ LIMITATIONS

- Single hardware stack (L4 / PRO6000 / H200) — results may not generalise to other GPU tiers or cloud inference endpoints.
- Benchmark scope covers 5 NLP task types; code generation, retrieval-augmented and long-context tasks are not included.
- EATS weights ($\beta = 0.60$, $\lambda = 0.40$) are fixed after calibration; sensitivity to alternative weight choices is not explored in this study.

SELECTED REFERENCES

[1] Hao et al. **Hybrid SLM–LLM Inference**, 2024 · [2] Oh & Kim, **Uncertainty-Aware Hybrid Inference**, 2025 · [3] Shao et al. **Division-of-Thoughts**, 2025 · [4] Kwon et al. **vLLM: PagedAttention**, SOSP 2023 · [5] Companion SLR (54 studies, PRISMA 2020) · [6] MMLU, ARC, GSM8K, HellaSwag, TruthfulQA.

Acknowledgement. We thank **HubX** for supporting this study with access to the virtual machine infrastructure used in our experiments.